

Health, safety and Environmental Management practises

UNIT-1

Health:

Health is a state of complete physical, mental and social well being and not only the absence of disease or infirmity. Health increases longevity and reduces mortality rate. physical exercise, balanced diet and hygiene are very crucial to maintain a good health.

Safety

Safety is a state in which hazards and conditions leading to physical, psychological or material harm are controlled in order to preserve the health and well-being of individuals and the community.

Environment

Environment can be defined as a sum total of all the living and non living elements and their effects that influence human life.

Environmental Management

Environmental Management is the practice of organizing human activities in order to limit their impact on the natural environment like protection of land, bodies of water and planet's atmosphere.

Basic Terms and their definitions

HSE

HSE stands for Health, safety and Environment. HSE encompasses a range of practices, policies and regulations to minimize hazards, prevent accidents and injuries and promote sustainable practices. HSE is vital for safeguarding the health and safety of individuals, protecting the environment and maintaining sustainable operations across different sectors.

Hazard:

A hazard is a process, phenomenon or human activity that may cause loss of life, injury or other health impacts, property damage, social and economic disruption or environmental degradation. Hazard is something that is dangerous and likely to cause a problem or damage. Ex: Earthquakes, Flood

HAZOP

HAZOP stands for Hazard and operability study and is a structured and systematic examination of a planned (or) existing process or operation in order to identify and evaluate problems that may represent risks to personnel or equipment.

PHA

PHA stands for process Hazard Analysis and it is ~~an~~ a process ^{or method} for identification of hazards. PHA is directed toward analyzing potential causes and consequences of fires, explosions, releases of toxic or flammable chemicals.

Accident

Accident is an undesirable or unfortunate happening that occurs unintentionally and usually results in harm, injury, damage or loss.

* Importance of safety:

1. Safety keeps us healthy:

If everyone is practicing safety in their work place by ~~and~~ keeping their surroundings and environment clean, they will not be getting sick and moreover they will be healthy.

2. Safety prevents Accidents and Injuries:

without safety procedures, bad things might happen at work or at home like gas leaks, fires etc. so it is important to practise safety in their work place or at homes to avoid accidents and injuries

3. Safety helps the environment:

using less energy means saving money on electricity bills and reducing greenhouse gas emissions. when everyone has a safe work place, they will be able to prevent many chemicals and other hazards from getting into the environment.

4. Safety supports our Economy:

By practising safety rules in ^{work place} place at work, employees can easily go about their work without accidents or injuries, thus the companies can be able to make money and meet the customer demands.

5. Safety provides peace of mind:

If a person knows that he/she is going to be in a safe place where accidents are not going to happen, then he/she will have peace of mind knowing that nothing bad will happen.

6. Safety prevents Diseases And Infections:

Without safety procedures, diseases and infections can be spread like Malaria, Dengue, cold, fever, AIDS/HIV & other life-threatening illnesses that people may come across the job. So it is important for every individual to practise safety procedures at work place.

7. Safety saves money:

When there are no injuries and accidents, we don't have to pay as much for medical expenses and thus can save money.

8. Safety teaches us to be prepared:

For people to be safe, they need to be prepared for any hazard that might happen. This means having the knowledge to deal with any type of emergency.

9. Safety encourages team work:

Teamwork is important in any workplace, and it is especially important when it comes to safety. When everyone is working together to keep themselves safe, it makes the job a lot easier.

10. Safety saves lives:

When everyone knows about these safety measures, then the number of accidents will be reduced, there will be no spreading of diseases and infections so that many lives will be saved. Hence safety is important.

Safety assurance :-

Safety assurance activities are at the core of your safety management system (SMS), with assurance functions monitoring compliance and performance as well as managing changes. Assurance consists of processes and activities undertaken to determine if your SMS is operating according to expectations and requirements.

Safety assurance includes systematic and ongoing monitoring and recording of your safety performance, as well as evaluating your safety management processes and practices. It also involves continuously monitoring your operating environment to detect changes which may introduce emerging risks or degrade any existing safety risk controls.

SMS Component	Elements
Safety assurance	Safety performance monitoring and measurement Management of Change Continuous improvement of the SMS

Safety assessment :-

A safety assessment is a systematic procedure that aims to identify anything endangering any people involved, including workers, contractors, visitors, customers, or the general public. The risk could also result in an otherwise unwanted outcome, including bodily harm, legal or regulatory liability, or loss of property or productivity.

Safety assurance :- assessment

what is a safety assessment?

→ Safety assessment is a vital part of your health and safety strategy.

- when you execute a proper assessment, you discover hazards and risks, and define,
- where control measures are needed to prevent illness & injury.
- This process will lead to a safety guide. Safety guides provide recommendations on how to comply with the safety requirements;
- they present good practices to help users strive to achieve high levels of safety through regulations.
- Safety or risk assessment is a thorough look at your workplace to identify situations, processes etc., that may cause harm, particularly to people.
- After identification, you analyse and evaluate the safety risks likely & severe.
- A safety assessment identifies potential damage and involves risk analysis and the likelihood of a risk.
- The overall purpose of safety assessment is to focus on providing your workers with a safe, health environment diminishing potential liability issues.
- Since each workplace is unique, your safety team should always customise the safety assessment based on individual conditions on-site.

Safety in Design and operation:-

Safety in design is a process that integrates hazard identification and risk assessment methods early in the design. The process considers how to eliminate, isolate or minimise the risks of death, injury and ill health to those who construct, operate, maintain, decommission or demolish an asset.

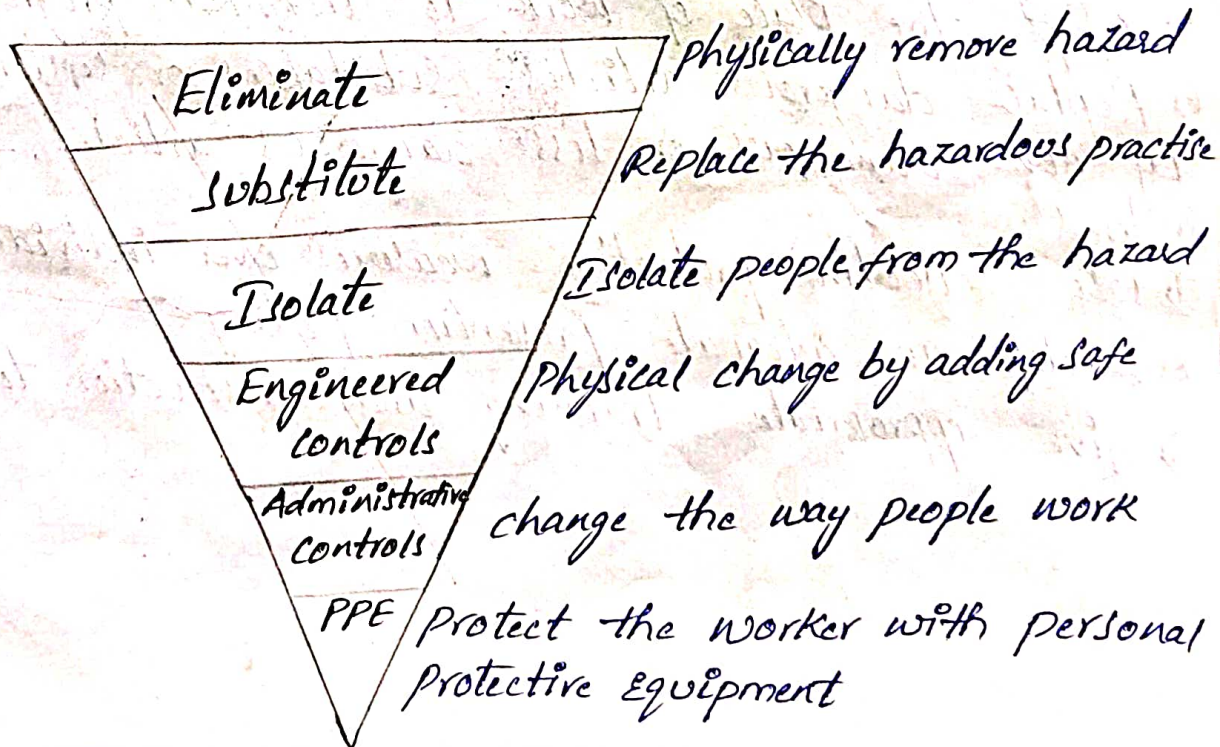
Safety in design begins at the conceptual and planning phases of a project. The emphasis is on making the right design choices as early as possible to enhance the safety of the project. These choices may include appropriate equipment selection, methods of construction, on-going maintenance provisions, or materials used.

Most construction safety risk mitigation is aimed at isolating, informing or controlling the hazard. The opportunity to consider the life cycle of the project and involve decision makers in the early design stages to eliminate a hazard is valuable. The earlier you can begin this process in design, the earlier it is to make changes that benefit everyone. Design development offers the greatest opportunity to incorporate improvements that can produce time and cost savings over the life of the asset. The key principle is that prevention is better than protection. This means that it is better to eliminate hazards than to try to manage them.

Greatest

Level of Control

Least



*The safety in design process helps designers fulfil their obligations by :

- providing a method to reduce the safety risk throughout the lifecycle of an asset.
- presenting an opportunity to challenge and improve designs through an open discussion.
- Enabling changes to be made early in the design stage that are most cost effective than retrofitted solutions made during operation and maintenance.
- Achieving better health and safety outcomes by considering and reconciling the interests of different parties involved with an asset.
- Outlining the key safety in design documentation requirements for each phase of the asset lifecycle.
- Continual improvement and knowledge sharing.

*The General principles of prevention are set out in descending order of preference as follows.

- 1) Avoid risks
- 2) Evaluate unavoidable risks.
- 3) Combat risks at source
- 4) Adapt work to the individual especially the design of places of work.
- 5) Adapt the place of work to technical progress.
- 6) Replace dangerous articles, substances or systems of work by non-dangerous or less dangerous articles, substances or systems.
- 7) Use collective protective measures over individual measures.
- 8) Develop an adequate prevention policy.
- 9) Give appropriate training and instruction to employees.

Organizing of Safety :-

Inferring from above general definitions, Safety organization can be defined as the structure and process by which groups of people are divided into sections or department. Each section or department is assigned specific safety function or duty. Authority and responsibility of everybody is clearly defined and interrelationship between them is specified for the accomplishment of organizational safety goals.

Need of organization :-

- facilitates administration and other functions of management process.
- facilitates growth and diversification.
- permits optimum use of technological improvements.
- Encourages use of human beings.
- Stimulates creativity.
- Attains maximum efficiency with minimum costs.

Nature of organization :-

- 1) It is a set up to realize the objectives
- 2) It is a co-operative activity
- 3) It is a process and structures both
- 4) Division into departments or groups. Division of labour.
- 5) Delegation of authority and responsibility. The chain of Superior Subordinate relationship is known as chain of Command.
- 6) Importance of human element
- 7) Vertical & horizontal relationship between man to man, Job to Job and department to department.
- 8) It is closely linked with planning. Other functions of management are not possible without organization.
- 9) It is flexible and accommodates environmental changes.
- 10) It is stable and resists challenges.
- 11) provides supervision, control and review. provides Mechanism of co-ordination.

- 12) Provides channels of communication.
- 13) Provides policies, procedures, rules and regulations for systematic functioning.
- 14) Backbone management and regenerates everything from beginning

Principles of organization :-

- Unity of objective - All are geared to common objectives
- Span of control - Each superior has limited subordinates to report him.
- Unity of command - Each subordinate has only one superior to get command.
- Scalar principle - clear chain of command from top to bottom.
- Co-ordination - Should exist among all individuals and groups.
- Communication should be open and clear.
- Facilitates leadership
- Flexibility - Should accommodate changes.
- Stability - Should resist challenges.
- Continuity - By training and development of executives and employees.
- Division of work - one member one function.
- Simplicity and clarity they add to efficacy of organization.

Types of organization :-

There are seven types of organization

- 1) Line, military, scalar or vertical
- 2) Line & staff organization
- 3) functional organization
- 4) product organization
- 5) project organization
- 6) Matrix organization
- 7) Committees.